

Circles & Coordinate Geometry

ANSWER KEY



The equation of a circle

- 1 Write the equation of a circle with center $(-4, 7)$ and radius 9.
 $(x + 4)^2 + (y - 7)^2 = 81$.
 - 2 The circle below has the radius shown. Find its area and circumference, in terms of π .
Area $= \pi(6)^2 = 36\pi$. Circumference $= 2\pi(6) = 12\pi$.
 - 3 A circle has diameter 26 and is centered at the origin. Write its equation, and state its radius.
Radius $= \frac{26}{2} = 13$. Equation: $x^2 + y^2 = 169$.
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Completing the square

- 4 Find the center and radius of $x^2 + y^2 - 6x + 8y - 11 = 0$.
 $(x^2 - 6x + 9) + (y^2 + 8y + 16) = 11 + 9 + 16 = 36 \Rightarrow (x - 3)^2 + (y + 4)^2 = 36$. Center $(3, -4)$, radius 6.
 - 5 Find the center and radius of $x^2 + y^2 + 10x - 4y + 13 = 0$.
 $(x^2 + 10x + 25) + (y^2 - 4y + 4) = -13 + 25 + 4 = 16 \Rightarrow (x + 5)^2 + (y - 2)^2 = 16$. Center $(-5, 2)$, radius 4.
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Distance, midpoint, and slope

- 6 Find the distance between $(-3, 5)$ and $(4, -7)$, to the nearest hundredth.
 $d = \sqrt{(4 - (-3))^2 + (-7 - 5)^2} = \sqrt{49 + 144} = \sqrt{193} \approx 13.89$.
 - 7 Find the midpoint and slope of the segment joining $(2, -6)$ and $(-8, 10)$.
Midpoint $= \left(\frac{2 - 8}{2}, \frac{-6 + 10}{2}\right) = (-3, 2)$. Slope $= \frac{10 - (-6)}{-8 - 2} = \frac{16}{-10} = -\frac{8}{5}$.
 - 8 Find the slope of the line through $(-2, 7)$ and $(6, -5)$, and state whether the line rises or falls from left to right.
Slope $= \frac{-5 - 7}{6 - (-2)} = \frac{-12}{8} = -\frac{3}{2}$. Since the slope is negative, the line falls from left to right.
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Coordinate proofs

- 9 Show that the triangle with vertices $A(0, 0)$, $B(4, 0)$, $C(4, 3)$ is a right triangle, using slopes.
Slope of $AB = \frac{0 - 0}{4 - 0} = 0$ (horizontal). Slope of BC is undefined (vertical, since $x = 4$ for both points). A horizontal segment and a vertical segment are perpendicular, so $\angle B = 90^\circ$, confirming a right triangle.

- 10 Show that $A(1, 2)$, $B(4, 3)$, $C(3, 6)$, $D(0, 5)$ form a parallelogram, by showing both pairs of opposite sides are parallel.
- Slope $AB = \frac{3-2}{4-1} = \frac{1}{3}$. Slope $DC = \frac{6-5}{3-0} = \frac{1}{3}$, so $AB \parallel DC$. Slope $AD = \frac{5-2}{0-1} = -3$. Slope $BC = \frac{6-3}{3-4} = -3$, so $AD \parallel BC$. Both pairs of opposite sides are parallel, so $ABCD$ is a parallelogram.
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Tangent lines to circles

- 11 A circle has center $O(2, 3)$ and radius 5. Point $P(2, 12)$ lies outside the circle. Find the length of the tangent from P to the circle, to the nearest hundredth.
- $OP = \sqrt{(2-2)^2 + (12-3)^2} = \sqrt{81} = 9$. Tangent length
 $= \sqrt{OP^2 - r^2} = \sqrt{81 - 25} = \sqrt{56} = 2\sqrt{14} \approx 7.48$.
- 12 Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$ on the circle.
- The radius to $(3, 4)$ has slope $\frac{4}{3}$, so the tangent (perpendicular to the radius) has slope $-\frac{3}{4}$. Line:
 $y - 4 = -\frac{3}{4}(x - 3)$, which simplifies to $3x + 4y = 25$.
- 13 A circle has center $(0, 0)$ and radius 10. Determine whether the point $(6, 9)$ lies inside, on, or outside the circle.
- Distance from center $= \sqrt{6^2 + 9^2} = \sqrt{117} \approx 10.82$, which is greater than the radius 10. The point lies outside the circle.
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Circle theorems

- 14 A central angle of 84° subtends an arc. Find the measure of the inscribed angle subtending the same arc.
- Inscribed angle $= \frac{1}{2}(84^\circ) = 42^\circ$.
- 15 A chord subtends the central angle shown below. Find the measure of the inscribed angle subtending the same (minor) arc.
- Inscribed angle $= \frac{1}{2}(100^\circ) = 50^\circ$.
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Arc length and sector area

- 16 A circle has radius 12 cm. Find the arc length subtended by a central angle of $\frac{5\pi}{6}$ radians.
- $s = r\theta = 12\left(\frac{5\pi}{6}\right) = 10\pi$ cm.
- 17 The circle below has the radius and shaded sector shown. Find the area of the sector, in terms of π .
- $72^\circ = \frac{2\pi}{5}$ radians. Sector area $= \frac{1}{2}r^2\theta = \frac{1}{2}(100)\left(\frac{2\pi}{5}\right) = 20\pi$.

- 18 A sector of a circle with radius 9 has area 27π . Find the central angle, in radians.

$$27\pi = \frac{1}{2}(81)\theta = 40.5\theta \Rightarrow \theta = \frac{27\pi}{40.5} = \frac{2\pi}{3}.$$

Conics: ellipses and hyperbolas

- 19 Identify the type of conic and state its center for $\frac{(x-2)^2}{16} + \frac{(y+1)^2}{9} = 1$.

An ellipse, centered at $(2, -1)$, with $a^2 = 16$ ($a = 4$, horizontal major axis) and $b^2 = 9$ ($b = 3$).

- 20 Identify the type of conic, its center, and its asymptote slopes for $\frac{(x+3)^2}{4} - \frac{y^2}{9} = 1$.

A hyperbola, centered at $(-3, 0)$, with $a = 2$, $b = 3$. Asymptote slopes: $\pm \frac{b}{a} = \pm \frac{3}{2}$.

A well-designed word problem: a Ferris wheel

- 21 This question has three parts. A circular Ferris wheel has a diameter of 40 m, with its center 22 m above the ground. Set up a coordinate system with the origin at ground level, directly below the center. (a) Write the equation of the circle traced by a passenger cabin. (b) A support cable runs from a ground anchor at $(26, 0)$ to a point of tangency on the wheel. Find the length of this cable, to the nearest hundredth of a meter. (c) Find the height above the ground of a cabin when it is 12 m horizontally from the vertical line through the center.

(a) Center $(0, 22)$, radius 20: $x^2 + (y - 22)^2 = 400$. (b) $OP = \sqrt{26^2 + 22^2} = \sqrt{1160} \approx 34.06$ m.

Tangent length $= \sqrt{OP^2 - r^2} = \sqrt{1160 - 400} = \sqrt{760} \approx 27.57$ m. (c) With $x = 12$:

$$12^2 + (y - 22)^2 = 400 \Rightarrow (y - 22)^2 = 256 \Rightarrow y = 22 \pm 16, \text{ so } y = 38 \text{ m or } y = 6 \text{ m}.$$